



AE4ASM505 NON-LINEAR MODELLING (USING F.E.M.)

Homework Assignment 1

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Contents

1	Problem A: Composite arc roof under point load	2
1.1	Task 1: Model	2
1.1.1	Geometry and Symmetry	2
1.1.2	Material and Section Properties	3
1.1.3	Boundary Conditions and Displacement Loading	4
1.1.4	Mesh	5
1.2	Task 2: Load-Displacement Curve	5
1.2.1	Riks Method	5
1.2.2	Newton-Raphson Method	5
1.2.3	Comparison of Numerical Methods	7
2	Problem B: Geometrically nonlinear truss structure	8
2.1	Task 1: Expressions for F_x and F_y	8
2.2	Task 2: Tangent Stiffness Matrix	10

1 Problem A: Composite arc roof under point load

1.1 Task 1: Model

1.1.1 Geometry and Symmetry

Taking advantage of the double symmetry of the structure, a reduced quarter-model was implemented to optimize computational efficiency. This simplification is justified by the existence of two primary planes of symmetry: the longitudinal YZ -plane located at the apex of the arc and the transverse XY -plane located at the mid-span of the roof.

The model represents the bottom-right quadrant of the panel. The geometry was created as a 3D deformable shell using an extrusion process. The base sketch, defined in the XY -plane and shown in Figure 1, consists of a circular arc with a radius of $R = 2500$ mm. This arc spans an angle of 6° , representing half of the total 12° opening angle of the full roof.

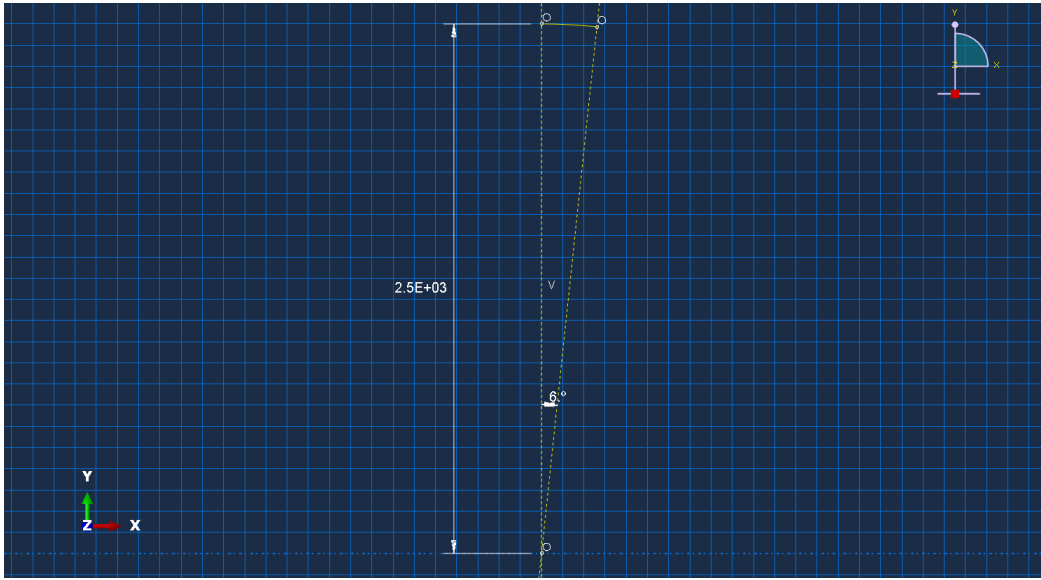


Figure 1: Circular arc sketch in the XY -plane ($R = 2500$ mm, $\theta = 6^\circ$).

To represent half of the total longitudinal span (500 mm), the sketch was extruded along the Z -axis to a depth of 250 mm. The resulting surface represents exactly one-fourth of the total arc roof, as shown in Figure 2.

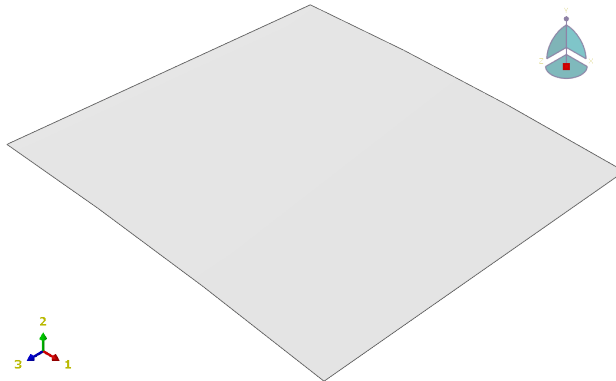


Figure 2: 3D shell part representing the quarter-model (bottom-right quadrant).

1.1.2 Material and Section Properties

The roof is composed of a $[0/90/0]$ composite laminate, where the physical 0° fibers are oriented along the global Z -axis. Each of the three plies has a thickness of 2 mm, resulting in a total shell thickness of 6 mm. The material properties were converted to a consistent MPa-based unit system (N, mm, MPa) as follows:

- $E_1 = 3300$ MPa, $E_2 = 1100$ MPa
- $G_{12} = G_{13} = 660$ MPa, $G_{23} = 500$ MPa
- $\nu_{12} = 0.25$

A **Composite Layup** was used in Abaqus to define the stacking sequence. To ensure the physical 0° fibers were correctly aligned with the global Z -axis (longitudinal direction) as specified in the problem, the material orientation was defined by assigning the global $(0, 0, 1)$ vector as the primary axis for the layup. This direct assignment aligns the local 1-direction of the shell elements with the Z -axis, allowing the $[0/90/0]$ sequence to be modeled exactly as defined physically. This orientation was verified using the material orientation plots shown in Figure 4 and 5.

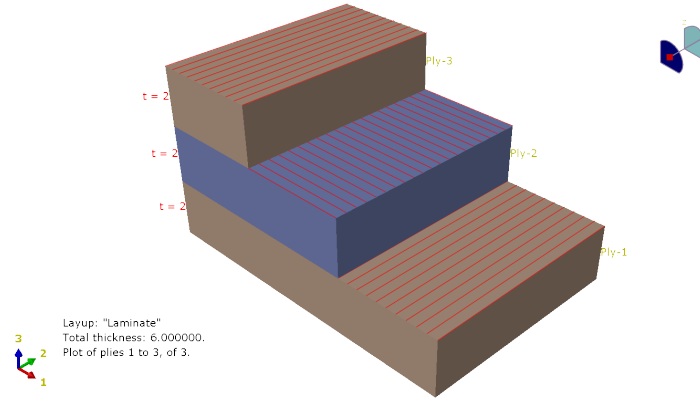


Figure 3: Ply stack plot.

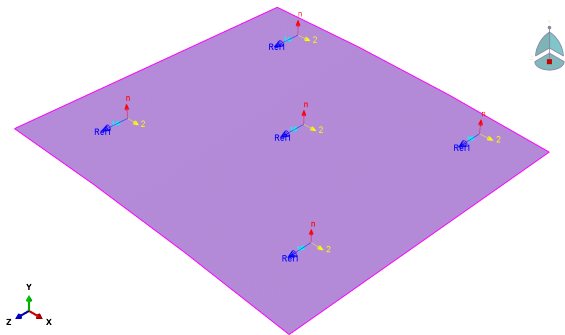


Figure 4: Material orientation for the 0° plies aligned with the Z -axis.

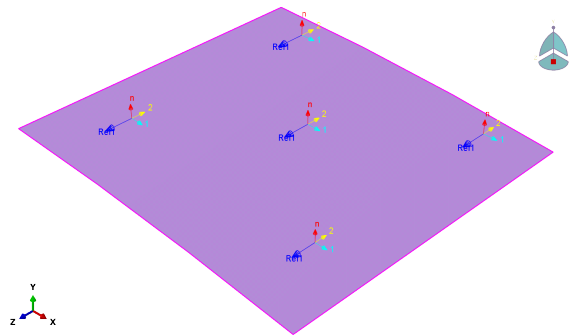


Figure 5: Material orientation for the 90° core ply.

1.1.3 Boundary Conditions and Displacement Loading

To accurately represent the full arc roof using a quarter-model, symmetry constraints were applied to the internal cut edges, while a hinged condition was applied to the physical support. These conditions are visualized in Figure 6 and defined as follows:

1. **Hinged Support (Outer straight edge):** This edge represents the physical connection of the roof. All translational degrees of freedom are restricted ($U_1 = U_2 = U_3 = 0$). To simulate a hinge, the rotation is restricted in all directions except for the rotation around the longitudinal axis of the support (UR_3 is free), allowing the arc to flex under load.
2. **Longitudinal Symmetry (Apex edge):** This edge lies on the YZ -plane of symmetry. An $XSMM$ boundary condition was applied, which restricts translation perpendicular to the plane ($U_1 = 0$) and rotations within the plane ($UR_2 = UR_3 = 0$). This ensures the two halves of the roof remain continuous at the peak.
3. **Transverse Symmetry (Mid-span edge):** This edge lies on the XY -plane of symmetry. A $ZSYMM$ boundary condition was applied, restricting longitudinal translation ($U_3 = 0$) and rotations $UR_1 = UR_2 = 0$. This effectively mirrors the model to represent the full 500 mm longitudinal span.

The external load was modeled using a displacement-control strategy applied to the vertex representing the center of the full roof. This node, located at the intersection of the two internal symmetry planes, was subjected to a prescribed vertical downward displacement of $U_2 = -30$ mm.

This approach was specifically selected to maintain numerical stability during the geometrically nonlinear phase of the analysis. Under traditional force-control, the solver would likely encounter a singularity at the limit point, leading to convergence failure as the structure loses its load-carrying capacity. By prescribing the displacement, the solver can capture the complete snap-through response. As shown in Figure 6, the displacement is applied as a boundary condition in the loading step, acting as the reference state for both the Newton-Raphson and Riks solvers.

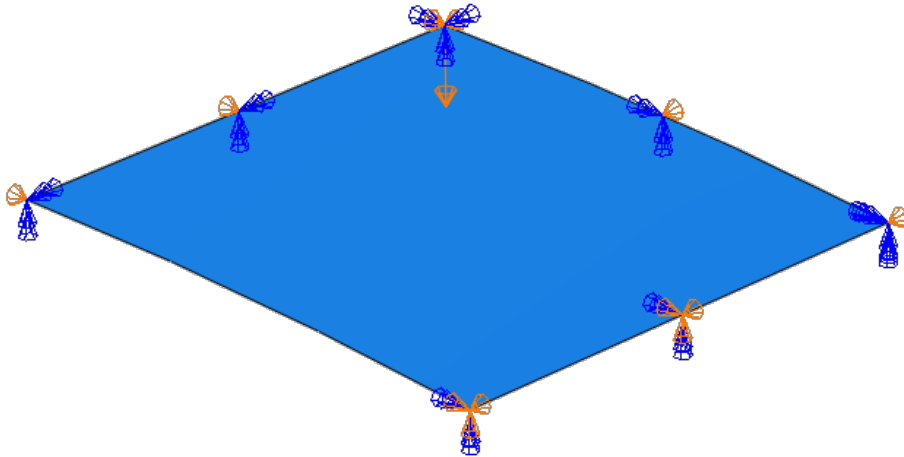


Figure 6: Definition of the displacement-controlled loading and boundary conditions in the Abaqus Step module.

1.1.4 Mesh

The quarter-model was discretized using a structured mesh technique with 4-node, doubly curved, thin shell elements with reduced integration (**S4R**). This element type is chosen for its ability to handle large displacements and rotations while avoiding shear locking in thin-shell applications. A mesh sensitivity study was performed, comparing 10×10 , 15×15 and 20×20 structured grids.

1.2 Task 2: Load-Displacement Curve

The vertical load-displacement relationship at the center of the arc roof was extracted up to a total downward displacement of 30 mm. The resulting curves represent the global response of the structure, where the reaction force values (RF_2) extracted from the quarter-model were multiplied by a factor of 4 to represent the total load (P) of the full structure.

1.2.1 Riks Method

The Riks method was implemented as the primary strategy to trace the equilibrium path, including the post-buckling snap-through and snap-back regions. A **Static, Riks** step was utilized where the -30 mm vertical displacement was set as the reference load. To ensure the analysis terminated precisely at the target depth, the stopping criteria were set to either a vertical displacement (U_2) of -30 mm at the loading node or a Load Proportionality Factor (LPF) of 1.0.

The incrementation was configured as **Automatic** with a maximum limit of 1000 increments to prevent premature termination. Both the initial and maximum arc length increments were restricted to 0.01, representing 1% of the reference state, while the minimum increment size was set to 1×10^{-5} to allow the solver to navigate high-curvature regions of the equilibrium path. Additionally, linear extrapolation of the previous state at the start of each increment was disabled to improve numerical stability during rapid stiffness changes.

As shown in Figure 7, the resulting curve successfully captures the complex "S-shaped" snap-through behavior. Notably, the equilibrium paths obtained using the 10×10 , 15×15 , and 20×20 mesh densities were found to be virtually identical, with only minor variations in the post-buckling snap-back region. This high degree of overlap confirms that the global structural response and the limit-point load have fully converged, demonstrating that even a coarser 10×10 mesh provides an accurate representation of the physical snap-through phenomenon when using the Riks algorithm.

1.2.2 Newton-Raphson Method

The second analysis was performed using the standard Newton-Raphson method within a **Static, General** step. This method also utilized the -30 mm displacement as a direct boundary condition. To maximize accuracy and robustness, the Full Newton solution technique was selected, which reforms the stiffness matrix at every iteration. Furthermore, linear extrapolation was explicitly disabled to prevent numerical overshooting at the onset of buckling.

The automatic incrementation scheme was configured with a maximum of 1000 increments. The initial and maximum increment sizes were both fixed at 0.01 to provide a smooth, detailed transition between the stable and unstable regions of the curve.

Numerical stability was found to be highly dependent on mesh refinement for this solver. The coarse 10×10 mesh failed to reach the target displacement, even after significantly reducing the minimum

increment size and increasing the maximum number of increments. This failure is attributed to the solver's inability to resolve the sharp snap-back instability present in the stiffer, coarse discretization. In contrast, as shown in Figure 8, both the 15×15 and 20×20 meshes successfully traced the equilibrium path. The curves for these two densities are almost perfectly coincident, confirming that the displacement-controlled Newton-Raphson method provides a converged response once a sufficient level of refinement is reached.

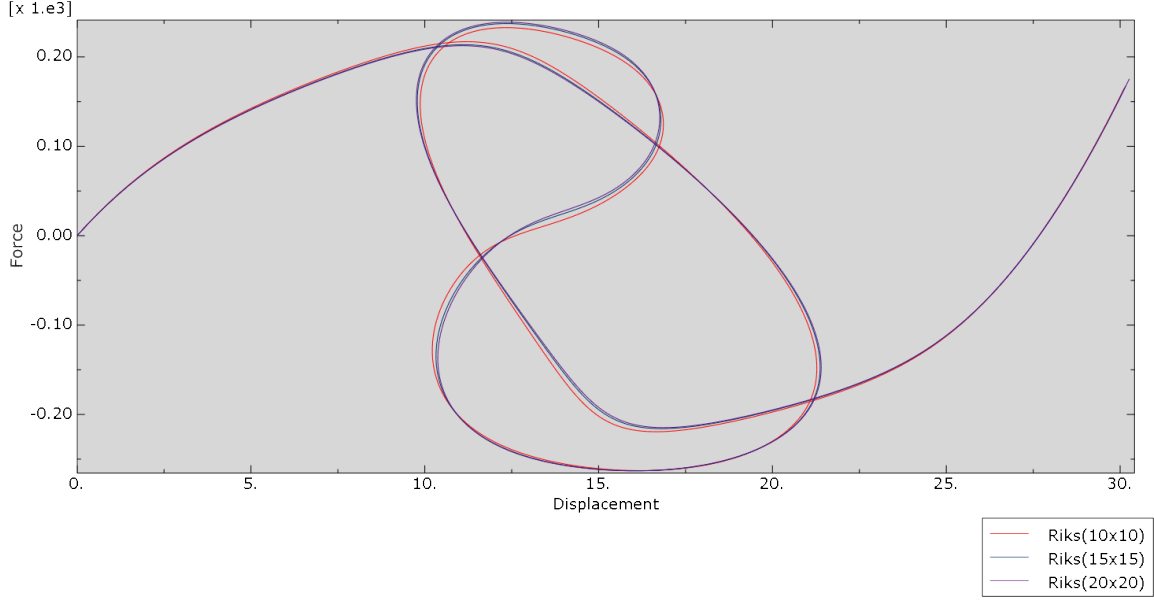


Figure 7: Load-displacement curves obtained using the Riks method across different mesh densities, demonstrating convergence.

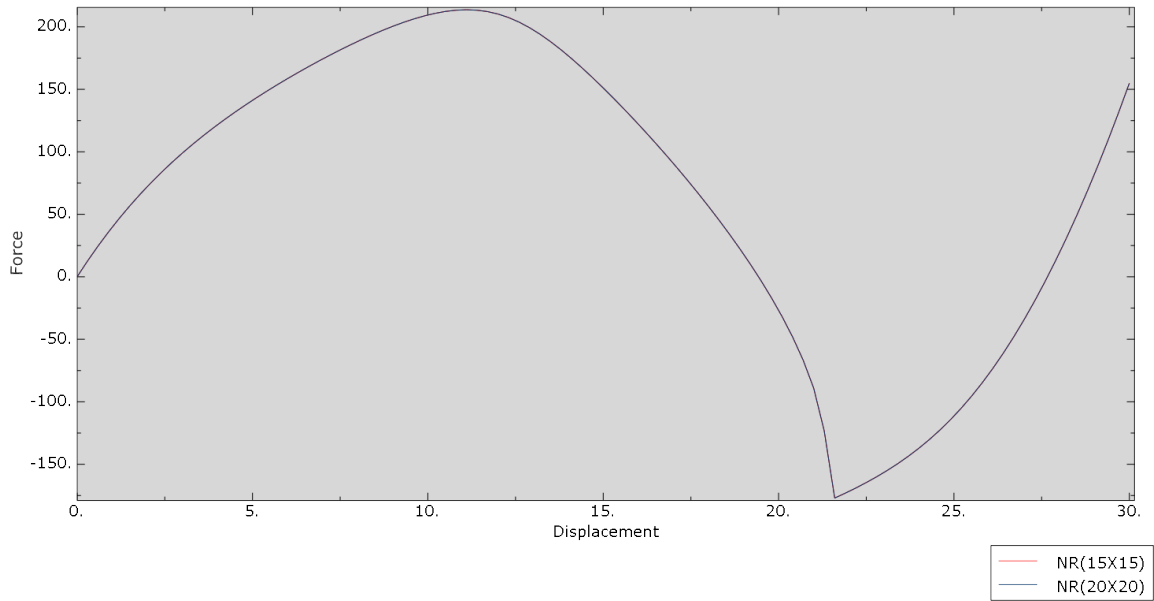


Figure 8: Load-displacement curves obtained using the Newton-Raphson method.

1.2.3 Comparison of Numerical Methods

The overlay of the load-displacement curves for the Riks and Newton-Raphson methods is presented in Figure 9. Both solvers exhibit a near-perfect overlap during the initial loading phase and the final stiffening stage once the arch is fully inverted. This agreement, combined with the fact that the 10×10 Riks results coincide with the 15×15 Newton-Raphson results, confirms that the global structural response is mesh-converged and independent of the algorithm used.

As a characteristic of thin and slender structures, the roof exhibits highly non-linear geometric behavior, including distinct limit points and unstable branches. The discretization must be sufficiently refined to capture these high-curvature gradients accurately. The Riks method provides the most complete representation of the structure's kinematics by tracing the full "S-shaped" equilibrium path, which includes the complex snap-back region where the displacement must physically reverse to maintain static equilibrium.

In contrast, the Newton-Raphson method successfully identifies the limit point at approximately 10.5 mm but is forced to follow a monotonic displacement path toward the -30 mm target due to the constraints of displacement control. Consequently, it cannot resolve the snap-back loop and instead exhibits a numerical jump between the stable pre-buckling and post-buckling states.

While the Riks method proved stable even at the coarsest 10×10 mesh, the Newton-Raphson solver required a minimum refinement of 15×15 to maintain numerical stability through the high-curvature gradients of the snap-through phase. Ultimately, Figure 9 demonstrates that while displacement-controlled Newton-Raphson is a robust tool for extracting the final inverted state, the Riks method is essential for capturing the complex, multi-valued equilibrium path characteristic of thin and slender composite arc structures.

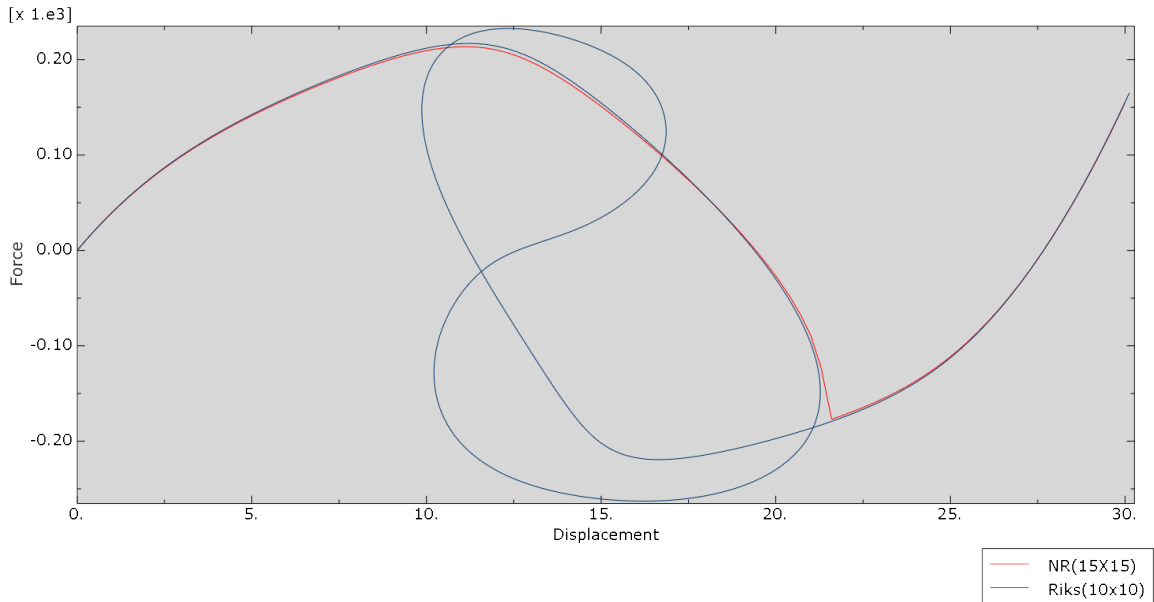


Figure 9: Final comparison between Riks and Newton-Raphson load-displacement curves.

$$L(w; F) = \text{MSE} + \alpha_1 \sum |w_j| + \alpha_2 \sum w_j^2$$

2 Problem B: Geometrically nonlinear truss structure

In this problem, a 2-member truss structure is analyzed using the Total Lagrangian Formulation. The structure consists of two truss members pinned at the bottom supports and connected at a free top nodeas illustrated in Figure 10.

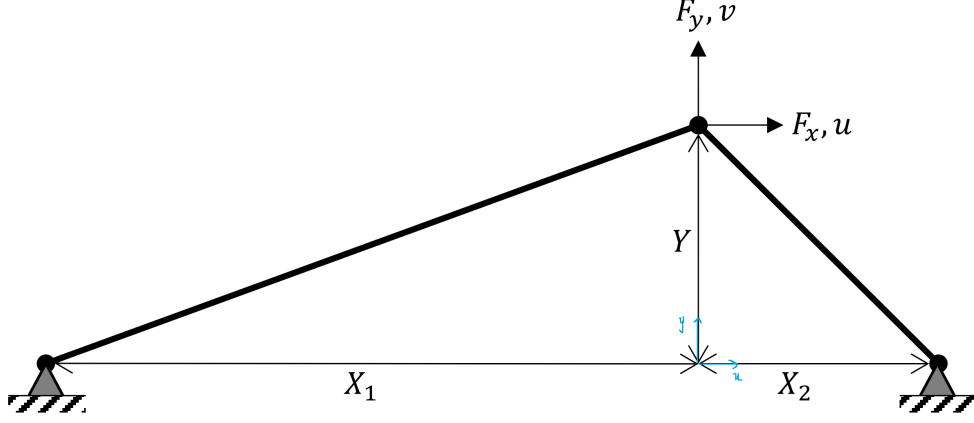


Figure 10: Schematic of the 2-member truss structure with the defined global coordinate system.

Let the left member be designated as Member 1 and the right member as Member 2. Based on the provided geometry, the initial coordinates of the nodes are:

- Node 1 (left support): $(-X_1, 0)$
- Node 2 (right support): $(X_2, 0)$
- Node 3 (top free node): $(0, Y)$

We explicitly define the vector of unknown degrees of freedom, $\mathbf{a}$, and the external force vector, $\mathbf{f}_{\text{ext}}$, applied at Node 3:

$$\mathbf{a} = \begin{bmatrix} u \\ v \end{bmatrix}, \quad \mathbf{f}_{\text{ext}} = \begin{bmatrix} F_x \\ F_y \end{bmatrix} \quad (2.1)$$

The initial lengths of the two members are $L_1 = \sqrt{X_1^2 + Y^2}$ and $L_2 = \sqrt{X_2^2 + Y^2}$.

2.1 Task 1: Expressions for F_x and F_y

According to the derivations from the class, the equilibrium of the structure is established using the weak form (Principle of Virtual Work) in the initial configuration. The internal force vector over the initial domain Ω_0 is defined as:

$$\mathbf{f}_{\text{int}} = \int_{\Omega_0} \mathbf{B}^T \mathbf{S} dV_0 \quad (2.2)$$

At equilibrium, the internal forces must balance the external forces ($\mathbf{f}_{\text{int}} = \mathbf{f}_{\text{ext}}$). For a linear elastic material, the 2nd Piola-Kirchhoff stress is $\mathbf{S} = \mathbf{C}\mathbf{E}$, where $\mathbf{C} = E$ is Young's modulus.

The Green-Lagrange strain E_{11} along the local axis is defined as:

$$E_{11} = \frac{\partial u_{\text{local}}}{\partial X} + \frac{1}{2} \left[\left(\frac{\partial u_{\text{local}}}{\partial X} \right)^2 + \left(\frac{\partial v_{\text{local}}}{\partial X} \right)^2 \right] \quad (2.3)$$

By projecting the global displacements (u, v) onto the local axes of each member, this differential expression simplifies to a function of initial relative coordinates $(\Delta X, \Delta Y)$ and relative displacements $(\Delta u, \Delta v)$:

$$E_{11} = \frac{1}{L^2} \left[\Delta X \Delta u + \Delta Y \Delta v + \frac{1}{2}((\Delta u)^2 + (\Delta v)^2) \right] \quad (2.4)$$

The variation of the strain gives the strain-displacement matrix $\mathbf{B}$:

$$\delta E_{11} = \frac{1}{L^2} \begin{bmatrix} \Delta X + \Delta u & \Delta Y + \Delta v \end{bmatrix} \begin{bmatrix} \delta u \\ \delta v \end{bmatrix} = \mathbf{B} \delta \mathbf{a} \quad (2.5)$$

For **Member 1** (Node 1 to Node 3): $\Delta X_1 = 0 - (-X_1) = X_1$, $\Delta Y_1 = Y - 0 = Y$, $\Delta u_1 = u - 0 = u$, and $\Delta v_1 = v - 0 = v$.

$$E_{11}^{(1)} = \frac{X_1 u + Y v + \frac{1}{2}(u^2 + v^2)}{L_1^2} \quad (2.6)$$

$$\mathbf{B}_1 = \frac{1}{L_1^2} \begin{bmatrix} X_1 + u & Y + v \end{bmatrix} \quad (2.7)$$

For **Member 2** (Node 2 to Node 3): $\Delta X_2 = 0 - X_2 = -X_2$, $\Delta Y_2 = Y - 0 = Y$, $\Delta u_2 = u - 0 = u$, and $\Delta v_2 = v - 0 = v$.

$$E_{11}^{(2)} = \frac{-X_2 u + Y v + \frac{1}{2}(u^2 + v^2)}{L_2^2} \quad (2.8)$$

$$\mathbf{B}_2 = \frac{1}{L_2^2} \begin{bmatrix} -X_2 + u & Y + v \end{bmatrix} \quad (2.9)$$

The total internal force vector is the sum of the contributions from both members integrated over their respective initial volumes ($V_0 = AL$):

$$\mathbf{f}_{\text{int}} = \int_{\Omega_{0(1)}} \mathbf{B}_1^T S_{11}^{(1)} dV_0 + \int_{\Omega_{0(2)}} \mathbf{B}_2^T S_{11}^{(2)} dV_0 = AL_1 \mathbf{B}_1^T (E E_{11}^{(1)}) + AL_2 \mathbf{B}_2^T (E E_{11}^{(2)}) \quad (2.10)$$

Expanding this equation provides the final expressions for the horizontal and vertical forces F_x and F_y :

$$F_x = \frac{EA}{L_1} E_{11}^{(1)} \left(\frac{X_1 + u}{L_1} \right) + \frac{EA}{L_2} E_{11}^{(2)} \left(\frac{-X_2 + u}{L_2} \right) \quad (2.11)$$

$$F_y = \frac{EA}{L_1} E_{11}^{(1)} \left(\frac{Y + v}{L_1} \right) + \frac{EA}{L_2} E_{11}^{(2)} \left(\frac{Y + v}{L_2} \right) \quad (2.12)$$

Substituting the full strain expressions into the force equations yields the exact functions of u and v :

$$F_x = \frac{EA}{L_1^3} \left[X_1 u + Y v + \frac{1}{2}(u^2 + v^2) \right] (X_1 + u) + \frac{EA}{L_2^3} \left[-X_2 u + Y v + \frac{1}{2}(u^2 + v^2) \right] (-X_2 + u) \quad (2.13)$$

$$F_y = \frac{EA}{L_1^3} \left[X_1 u + Y v + \frac{1}{2}(u^2 + v^2) \right] (Y + v) + \frac{EA}{L_2^3} \left[-X_2 u + Y v + \frac{1}{2}(u^2 + v^2) \right] (Y + v) \quad (2.14)$$

2.2 Task 2: Tangent Stiffness Matrix

The tangent stiffness matrix $\mathbf{K}$ is the derivative of the internal force vector with respect to the vector of unknowns $\mathbf{a}$:

$$\mathbf{K} = \frac{\partial \mathbf{f}_{\text{int}}}{\partial \mathbf{a}} = \int_{\Omega_0} \frac{\partial \mathbf{B}^T}{\partial \mathbf{a}} \mathbf{S} dV_0 + \int_{\Omega_0} \mathbf{B}^T \frac{\partial \mathbf{S}}{\partial \mathbf{a}} dV_0 \quad (2.15)$$

This equation naturally separates the tangent stiffness matrix into two distinct components: the geometric stiffness matrix ($\mathbf{K}^{\text{Geo}}$) and the material stiffness matrix ($\mathbf{K}^{\text{Mat}}$). Substituting the constitutive relationship $\partial \mathbf{S} / \partial \mathbf{a} = \mathbb{C} \mathbf{B}$ (where $\mathbb{C} = E$ for our 1D truss members), we obtain:

$$\mathbf{K} = \mathbf{K}^{\text{Geo}} + \mathbf{K}^{\text{Mat}} \quad (2.16)$$

where

$$\mathbf{K}^{\text{Geo}} = \int_{\Omega_0} \frac{\partial \mathbf{B}^T}{\partial \mathbf{a}} \mathbf{S} dV_0 \quad \text{and} \quad \mathbf{K}^{\text{Mat}} = \int_{\Omega_0} \mathbf{B}^T \mathbb{C} \mathbf{B} dV_0 \quad (2.17)$$

For our structure, the total stiffness matrix is the sum of the contributions from the two truss members: $\mathbf{K} = \mathbf{K}^{(1)} + \mathbf{K}^{(2)}$.

Starting with **Member 1**, we construct the material stiffness component using $\mathbf{B}_1$ derived in Task 1:

$$\mathbf{K}^{\text{Mat}(1)} = \int_{\Omega_{0(1)}} \mathbf{B}_1^T E \mathbf{B}_1 dV_0 = AL_1 (\mathbf{B}_1^T E \mathbf{B}_1) = \frac{EA}{L_1^3} \begin{bmatrix} (X_1 + u)^2 & (X_1 + u)(Y + v) \\ (X_1 + u)(Y + v) & (Y + v)^2 \end{bmatrix} \quad (2.18)$$

To find the geometric stiffness, we take the derivative of $\mathbf{B}_1^T$ with respect to $\mathbf{a}$:

$$\mathbf{B}_1^T = \frac{1}{L_1^2} \begin{bmatrix} X_1 + u \\ Y + v \end{bmatrix} \implies \frac{\partial \mathbf{B}_1^T}{\partial \mathbf{a}} = \frac{1}{L_1^2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (2.19)$$

Substituting this into the geometric stiffness integral:

$$\mathbf{K}^{\text{Geo}(1)} = \int_{\Omega_{0(1)}} \frac{\partial \mathbf{B}_1^T}{\partial \mathbf{a}} S_{11}^{(1)} dV_0 = AL_1 \left(\frac{1}{L_1^2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) S_{11}^{(1)} = \frac{AE}{L_1} E_{11}^{(1)} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (2.20)$$

Applying the exact same procedure to **Member 2**:

$$\mathbf{K}^{\text{Mat}(2)} = \int_{\Omega_{0(2)}} \mathbf{B}_2^T E \mathbf{B}_2 dV_0 = AL_2 (\mathbf{B}_2^T E \mathbf{B}_2) = \frac{EA}{L_2^3} \begin{bmatrix} (-X_2 + u)^2 & (-X_2 + u)(Y + v) \\ (-X_2 + u)(Y + v) & (Y + v)^2 \end{bmatrix} \quad (2.21)$$

$$\mathbf{K}^{\text{Geo}(2)} = \int_{\Omega_{0(2)}} \frac{\partial \mathbf{B}_2^T}{\partial \mathbf{a}} S_{11}^{(2)} dV_0 = AL_2 \left(\frac{1}{L_2^2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) S_{11}^{(2)} = \frac{AE}{L_2} E_{11}^{(2)} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (2.22)$$

The total 2×2 tangent stiffness matrix for the structure is the direct assembly of the individual material and geometric stiffness matrices:

$$\mathbf{K} = \mathbf{K}^{\text{Mat}(1)} + \mathbf{K}^{\text{Mat}(2)} + \mathbf{K}^{\text{Geo}(1)} + \mathbf{K}^{\text{Geo}(2)} \quad (2.23)$$

This can be explicitly written as $\mathbf{K} = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix}$, where the individual terms are:

$$K_{11} = \frac{EA}{L_1^3} (X_1 + u)^2 + \frac{EA}{L_2^3} (-X_2 + u)^2 + \frac{EA}{L_1} E_{11}^{(1)} + \frac{EA}{L_2} E_{11}^{(2)} \quad (2.24)$$

$$K_{12} = K_{21} = \frac{EA}{L_1^3}(X_1 + u)(Y + v) + \frac{EA}{L_2^3}(-X_2 + u)(Y + v) \quad (2.25)$$

$$K_{22} = \frac{EA}{L_1^3}(Y + v)^2 + \frac{EA}{L_2^3}(Y + v)^2 + \frac{EA}{L_1}E_{11}^{(1)} + \frac{EA}{L_2}E_{11}^{(2)} \quad (2.26)$$

where $E_{11}^{(1)}$ and $E_{11}^{(2)}$ are the Green-Lagrange strain expressions defined in Task 1.

References

- [1] Dr. Boyang Chen, *Lecture Notes - Non-Linear Modelling (using F.E.M.)*, Brightspace.